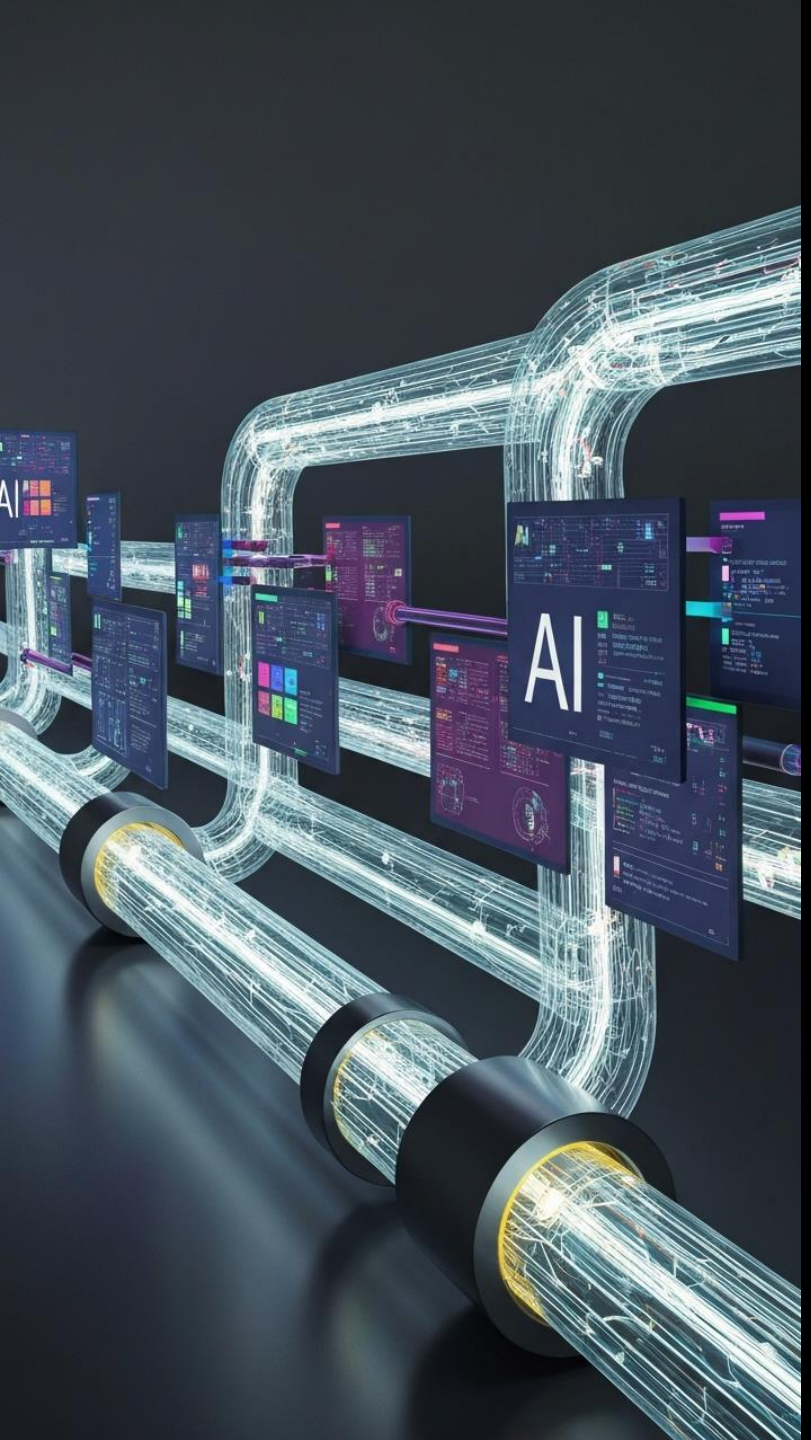


STAR EAST
A TECHWELL EVENT

DevOps-Driven Quality Engineering with AI

Turning Pipelines into Quality Platforms

April 26 – May 1, 2026
Orlando, FL

About Me

- DevOps Engineering Manager at KinderCare Learning Companies.
- 20 years in the software industry.
- Design reliable, secure, and cost-optimized data and AI platforms.
- Drives enterprise AI adoption through a governed operating model that accelerates time-to-market while reducing risk and spend.
- Expertise spans BI strategy, data platform architecture, MLOps, observability, and value realization.
- Deep technical rigor and business expertise in DevOps and Reliability Engineering.

The Traditional Quality Bottleneck

Quality Engineering: A Gatekeeper or a Continuous Enabler?

Common perception: QE as final hurdle.

Quality Engineering is often seen as a checkpoint that must be passed before release.

DevOps focus: Speed and efficiency.

DevOps emphasizes rapid delivery and streamlined workflows.

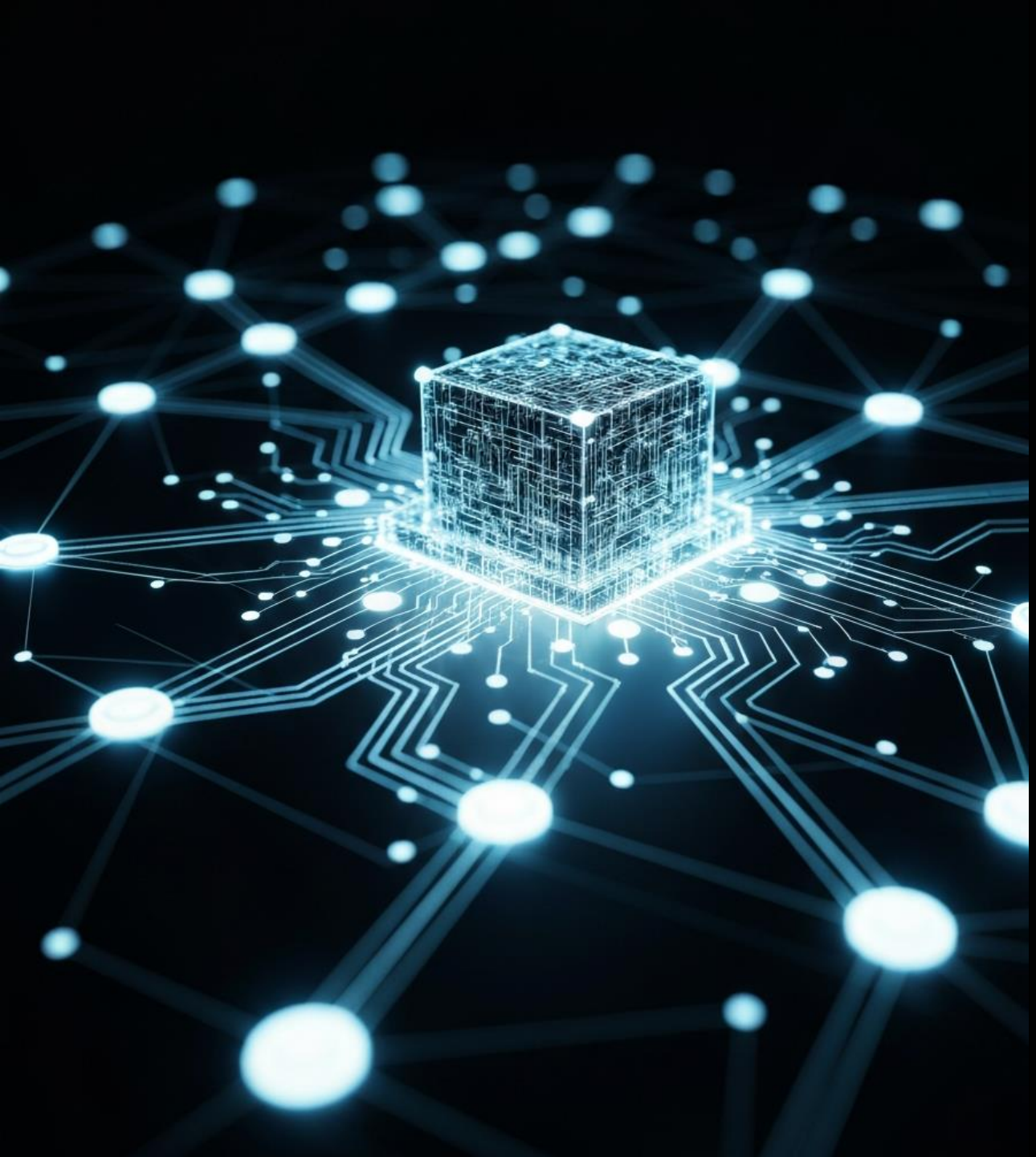
The inherent conflict: Speed vs. Thoroughness.

The drive for speed can clash with the need for comprehensive quality assurance.

Key Considerations

The Ideal State: Quality and Flow

- Achieving rapid delivery without compromising quality.
- The synergy between DevOps principles and robust QE.
- How can we make quality an integral part of the pipeline?



AI: The Game Changer for Proactive Quality

Leveraging AI as a catalyst to move beyond reactive testing, enhance signaling, and enable proactive issue identification and resolution.

Context

Our Journey: Reframing QE

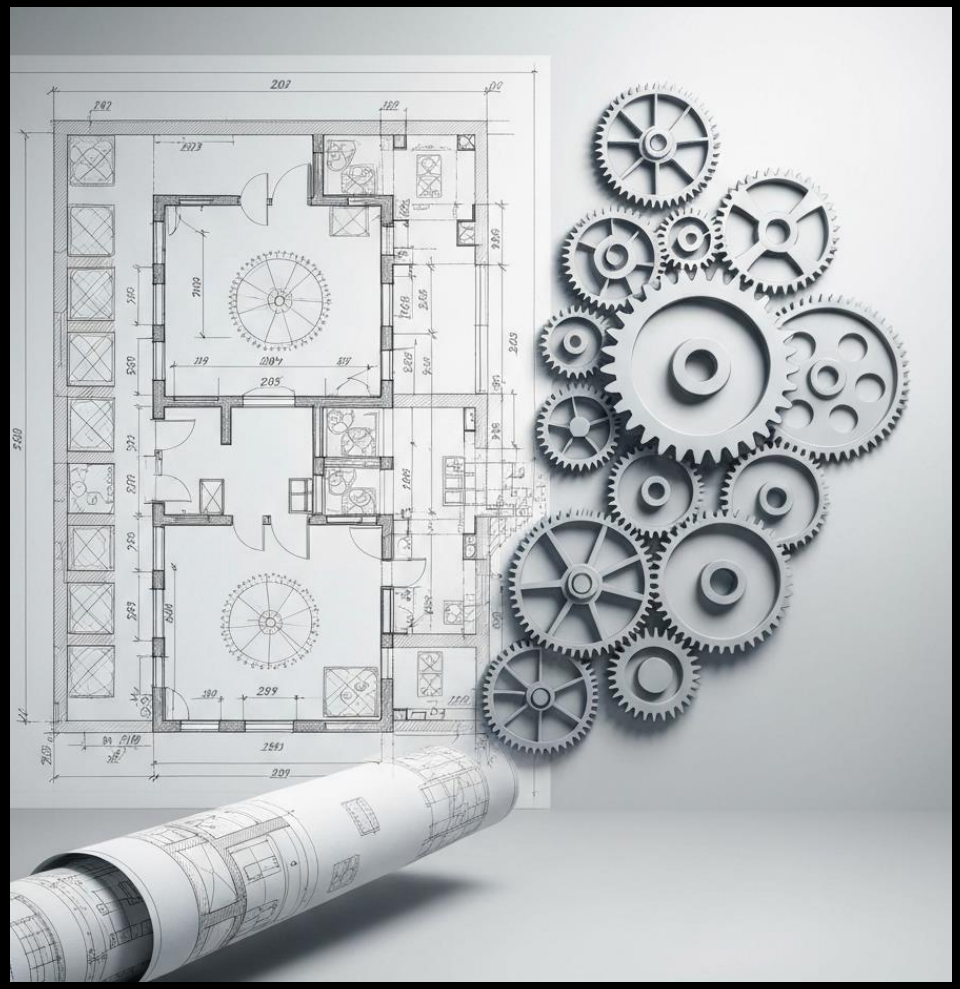
Any organization domain.

Transforming QE into an embedded platform within DevOps.

The shift towards a quality-first, integrated approach.

Foundation: SLOs to Executable Checks

Translating Business Goals into Testable Actions



- **Defining Service Level Objectives (SLOs)**
  Establish clear, measurable objectives for service performance.
- **Mapping SLOs to Executable Checks**
  Translate high-level SLOs into specific, automated quality tests.
- **Aligning Quality Metrics with User Experience**
  Ensure that performance metrics directly reflect and improve the end-user experience.

Leveraging Production Telemetry

Data-Driven Insights for Quality

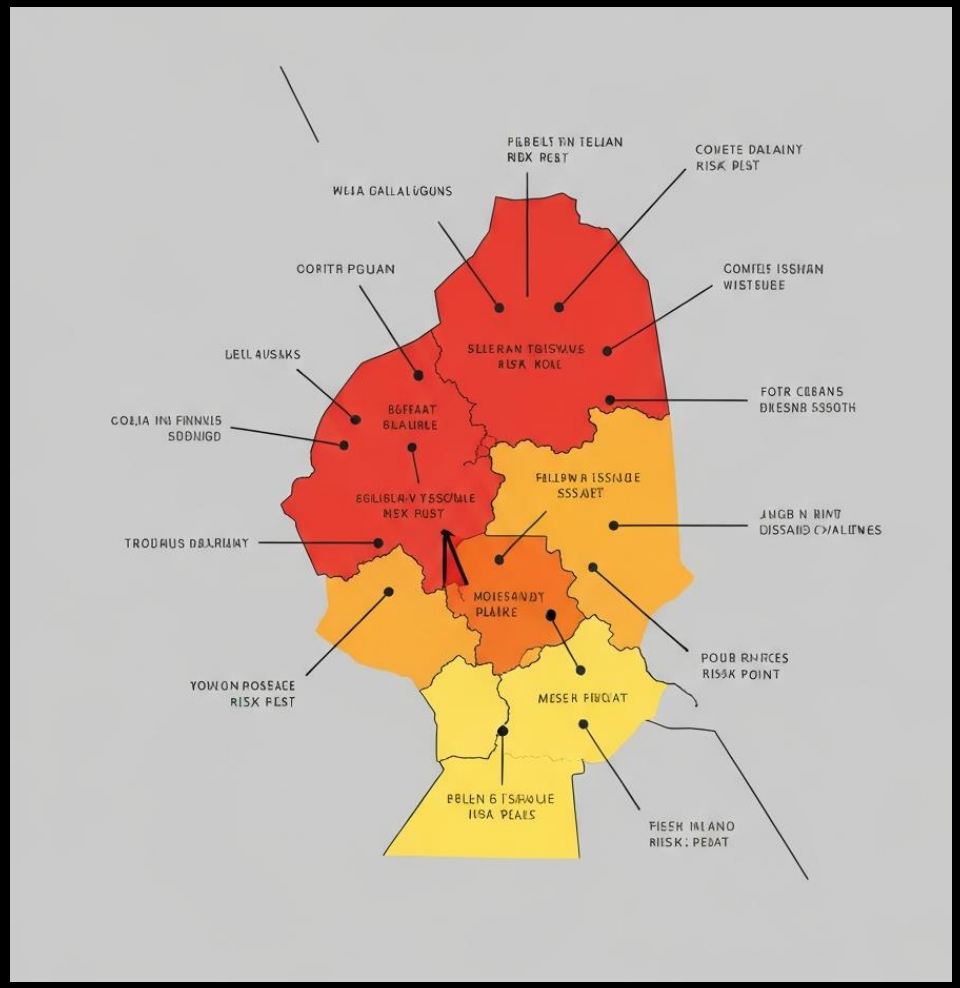
Analyzing logs, metrics, and traces.

Extracting actionable information from production data.

Understanding real-world system behavior.

Creating a Risk Map

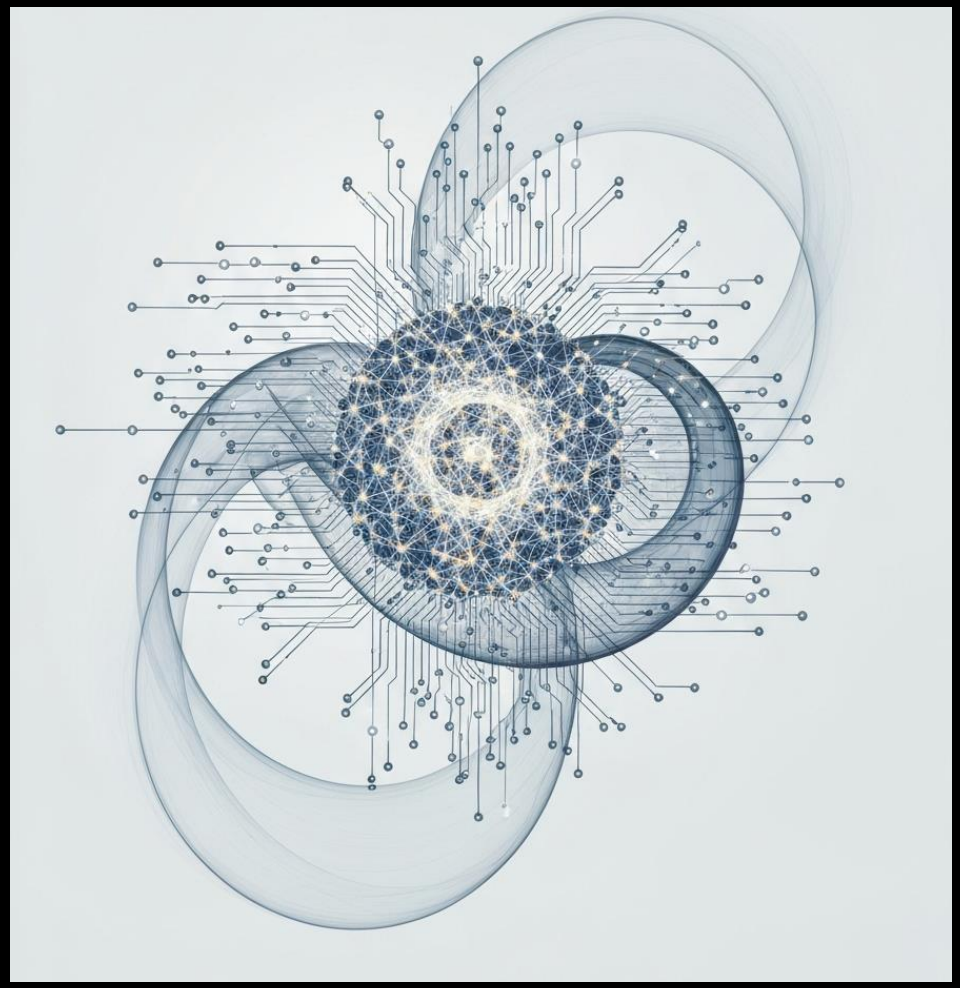
Prioritizing Quality Efforts with Precision



- **Identifying high-risk areas based on telemetry.**
  Pinpoint critical areas needing attention through data analysis.
- **Directing testing efforts where they matter most.**
  Focus your testing on the most impactful functionalities.
- **Maximizing the impact of limited testing resources.**
  Ensure your resources are used efficiently for the greatest return.

AI in Action: Test Generation

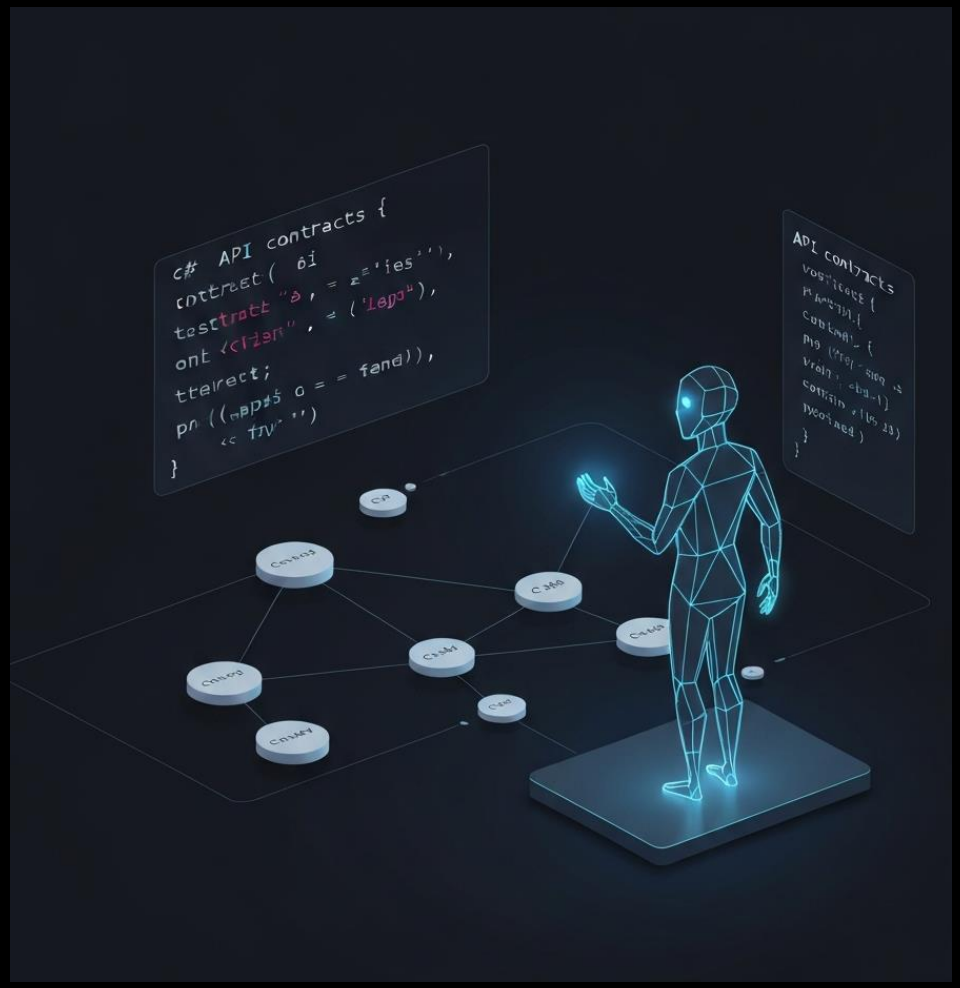
Automating the Creation of Test Artifacts



- **AI-powered generation of test scenarios.**
  Leveraging artificial intelligence to automatically create diverse and relevant test scenarios.
- **Dynamic creation of comprehensive test data.**
  Generating test data that adapts to various testing needs, ensuring thoroughness.
- **Reducing manual effort and increasing test coverage.**
  Streamlining the testing process by minimizing manual tasks and expanding the scope of tests.

AI in Action: Contract Testing

Ensuring Seamless Service Interactions



- **AI-Powered Contract Generation**
  AI assistance in generating and maintaining API contracts.
- **Proactive Issue Prevention**
  Preventing integration issues before they arise.
- **System-Wide Consistency**
  Maintaining consistency across distributed systems.

AI-Driven Test Suite Prioritization

Optimizing Test Execution for Maximum Value

- **Prioritizing test suites based on risk and user impact.**
  Focuses on the most crucial tests to ensure quality where it matters most.
- **Running the most critical tests first.**
  Ensures that high-impact issues are identified early in the development cycle.
- **Reducing feedback loops and accelerating delivery.**
  Enables quicker iterations and faster time-to-market by streamlining the testing process.

Key Capabilities

AI for Anomaly Detection

- Proactive detection of anomalies in logs, traces, and metrics.
- Early warning system for potential issues.
- Identifying deviations from normal behavior.

From Red Builds to Actionable Bugs

AI-Powered Failure Analysis

- **Identifying patterns for failure clustering.**
  Leveraging AI to group similar build failures allows for more efficient analysis and quicker resolution.
- **AI-generated root-cause analysis summaries.**
  Receive concise summaries pinpointing the origin of build failures, saving valuable debugging time.
- **Transforming build failures into clear, actionable bug reports.**
  Automatically convert complex failure data into well-defined bug tickets that development teams can immediately address.

Ensuring Safe AI Adoption

Governance and Responsible AI Implementation

- **Establishing robust governance frameworks.**
  Define clear guidelines and oversight mechanisms for AI development and deployment.
- **Implementing prompt templates as code for consistency.**
  Standardize AI interactions through codified prompt templates to ensure predictable and reliable outputs.
- **Mitigating risks associated with AI usage.**
  Proactively identify and address potential threats and ethical concerns arising from AI implementation.

Handling Sensitive Data with AI

PII Scrubbing and Data Privacy

AI techniques for identifying and redacting Personally Identifiable Information (PII).

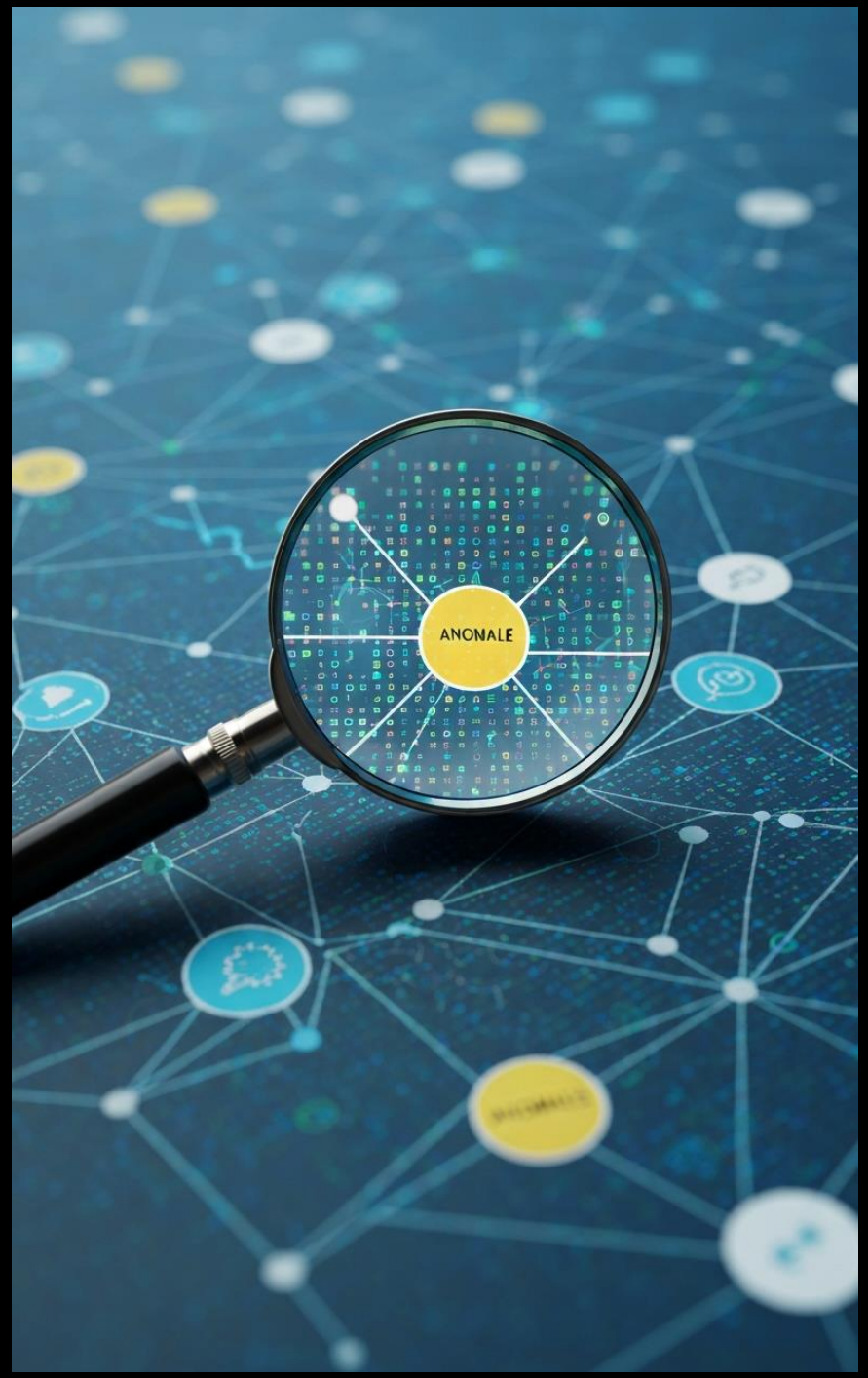
Leveraging artificial intelligence to accurately detect and remove sensitive personal data.

Ensuring data privacy and compliance.

Maintaining adherence to privacy regulations and standards.

Protecting user data throughout the testing process.

Safeguarding sensitive information during all stages of testing.



Key Strategies

Maintaining Data Integrity with AI

Utilizing AI to manage and validate golden datasets.

Detecting and addressing data drift over time.

Ensuring test data remains relevant and accurate.

The Repeatable Playbook

Your Guide to DevOps-Driven QE with AI

Key Areas of Focus

Playbook Component 1: SLO-to-Tests Mapping

- Step-by-step guide for defining and implementing SLO-driven testing.
- Tools and techniques for effective mapping.
- Measuring success through SLO achievement.

Playbook Component 2: AI-Assisted Triage

Streamlining Incident and Failure Resolution

- **Workflow for leveraging AI in triaging issues.**
  This outlines the process of integrating AI to automate and optimize the initial assessment of incidents and failures.
- **Reducing Mean Time To Resolution (MTTR).**
  Focuses on how AI-powered triage contributes to a faster overall resolution time for incidents.
- **Empowering debugging and faster fixes.**
  Explains how AI assistance aids in identifying root causes and expedites the implementation of solutions.

Playbook Component 3: Cost-Aware Environments

Optimizing Infrastructure for Efficiency

- **Cost-Effective Test Environment Management**
  Patterns for managing test environments cost-effectively.
- **Resource Utilization Balance**
  Balancing resource utilization with testing needs.
- **Sustainable & Scalable Infrastructure**
  Sustainable and scalable test infrastructure.

The Ultimate Outcome: Release Readiness Score

Connecting Quality to Business Value

- **A single, unified score for release readiness.**
  Provides a consolidated view of your release's preparedness.
- **Directly linking quality metrics to business outcomes.**
  Understand the business impact of your quality efforts.
- **Data-driven decisions for go/no-go releases.**
  Make informed release decisions based on objective data.

Accelerating Delivery with Confidence

The Benefits of a DevOps-Driven Quality Platform

- **Faster release cycles.**
  Streamline your development and deployment processes for quicker product delivery.
- **Higher product quality and stability.**
  Ensure robust and reliable software through continuous integration and testing.
- **Improved customer satisfaction.**
  Deliver exceptional experiences that meet and exceed user expectations.

Key Takeaways

Summarizing the Path Forward

QE as a platform, not a gate.

Shift focus from restrictive gatekeeping to enabling a robust quality platform.

AI as an enabler of proactive quality.

Leverage Artificial Intelligence to anticipate and prevent quality issues before they arise.

The power of SLOs and telemetry.

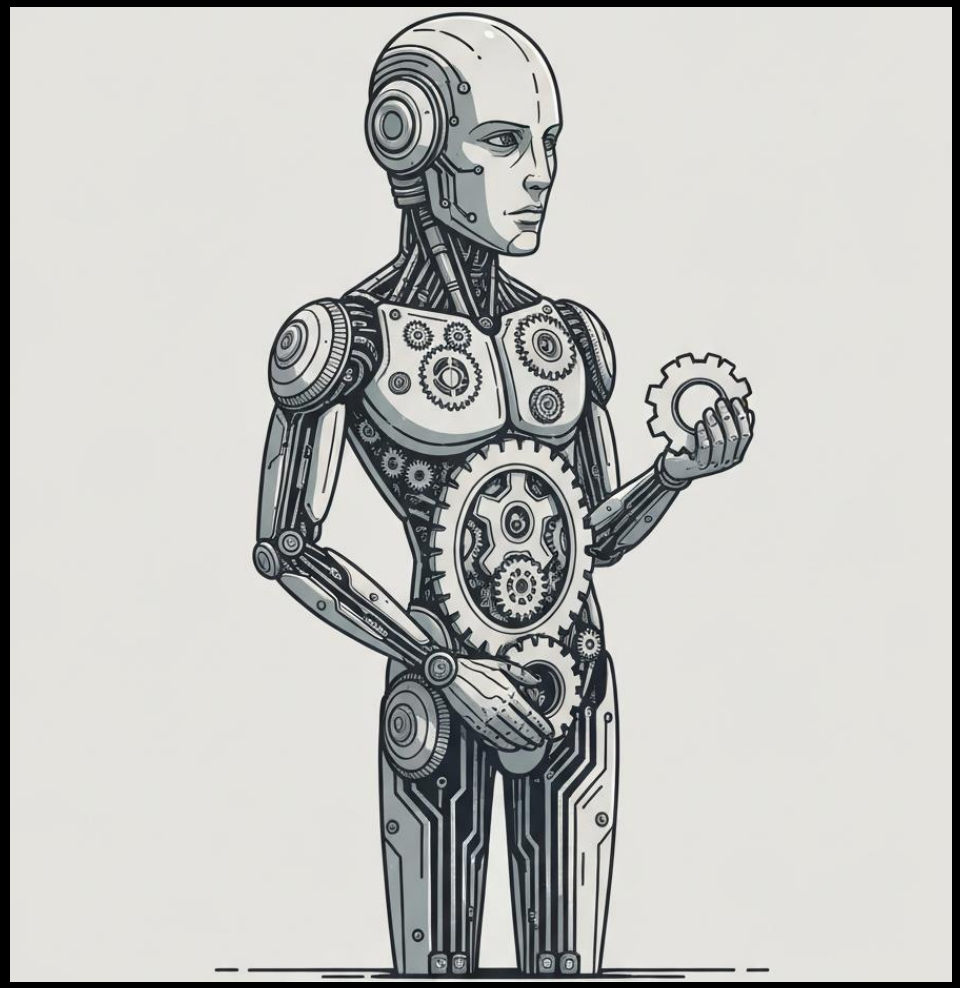
Utilize Service Level Objectives (SLOs) and telemetry data for effective monitoring and performance.

A focus on actionable insights and safe AI.

Prioritize generating practical insights from data and ensuring the responsible and secure implementation of AI.

The Future of Quality Engineering

Continuous Evolution and Innovation



- **Embracing AI and automation.**
  Leveraging artificial intelligence and automated processes to enhance quality assurance.
- **Building intelligent, adaptive quality systems.**
  Developing quality systems that can learn and adjust to changing development environments.
- **Staying ahead in a fast-paced development world.**
  Ensuring quality practices keep pace with rapid software development cycles.

Open Floor for Questions & Discussion

Let's Discuss Your Quality Challenges

Thank You!

Shalini Sudarsan

Shalini Sudarsan

Engineering Leader | Driving Business-Aligned Tech Solutions | Site Reliability Engineering/DevSecOps |...